nature COMMUNICATIONS

Perturbation-response genes reveal signaling footprints in cancer gene expression

Michael Schubert[1], Bertram Klinger[2,3], Martina Klünemann[2,3], Anja Sieber[2,3], Florian Uhlitz[2,3], Sascha Sauer[4], Mathew J. Garnett[5], Nils Blüthgen[2,3] & Julio Saez-Rodriguez[1,6]

HP-F2: Part 3—Bioinformatics

Presenter: Yilu Wang
Major Cancer Biology
04.12.2025

UNIVERSITÄT HEIDELBERG
ZUKUNFT SEIT 1386

dkfz. GERMAN CANCER RESEARCH CENTER IN THE HELMHOLTZ ASSOCIATION
Research for a Life without Cancer

Authors

Michael Schubert

2018: PhD student at **EMBL–EBI** (Cambridge, UK)

2025: Assistant Professor at the Medical University of Innsbruck

Julio Saez-Rodriguez

2018: Group leader at **EMBL-EBI** (Cambridge, UK) and RWTH Aachen University

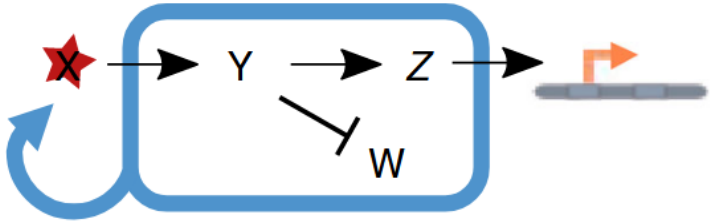
2025: Head of Research at EMBL-EBI

EBI: European Bioinformatics Institute

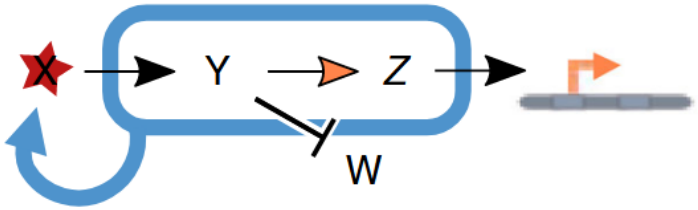
Rationale: Reading pathway activity from gene expression

- Cancer can arise from aberrant **cell signaling**, which is a focal point of drug research
- Inferring signaling pathway activity from **gene expression** is a common approach
- Quantifying **pathway member transcripts** disregards the effect of **post-translational modifications**

GO, pathway enrichment

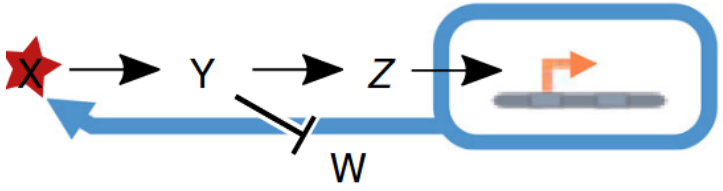


SPIA, Pathifier, PARADIGM



- Perturbation experiments addresses this by measuring the **downstream responses** → **pathway "footprints"/signatures**

Signatures, PROGENy: Pathway RespOnsive GENes



GO: Gene Ontology
SPIA: Signaling Pathway Impact Analysis

Why PROGENy?

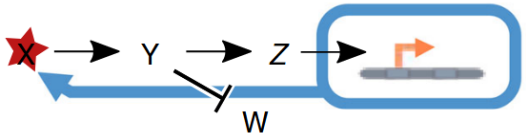
Methodology

Idea:

Infer pathway activity from **downstream signatures**, not pathway member mRNAs

Pipeline:

1. Perturbation → response genes (PRGs)



2. Learning signatures
3. Scoring new samples

Why this works:

Regards the effect of **post-translational modifications**

Data

Training set:

A large compendium of publicly available **perturbation experiments** across tissues and cell lines

Perturbation-learned signatures turn complex expression into actionable **pathway activity**— better linked to **mutations, drugs, and patient survival**

PROGENy uses downstream gene signatures to reflect pathway activity

Data set of experiments → Expression contrast → Z-scores → Linear regression model → Coefficient matrix → Pathway scores

Pathway name	Size of the data set
EGFR	106
MAPK	88
TNFa	69
Hypoxia	66
JAK-STAT	66
NFkB	46
VEGF	36
TGFb	31
PI3K	27
p53	23
Trail	10

Analyze > 500 perturbation experiments across pathways

For each gene:

For gene X:

$$Z = P \times W$$

Gene X (Experiments)	11 pathways (Experiments)			Gene X (Pathways)
$z_{x,1}$	1	…	1	$w_{x,1}$
$z_{x,2}$	1	…	0	$w_{x,2}$
…	…	…	…	…
$z_{x,n}$	0	…	0	$w_{x,11}$

n is the number of experiment

1 for activation, -1 for inhibition, otherwise 0

$1 \times n$ Z-score

$11 \times n$ Pathway perturbation

1×11 Weight

Fit for the weights for 11 pathways **per gene** → identify ~100 PRGs **per pathway**

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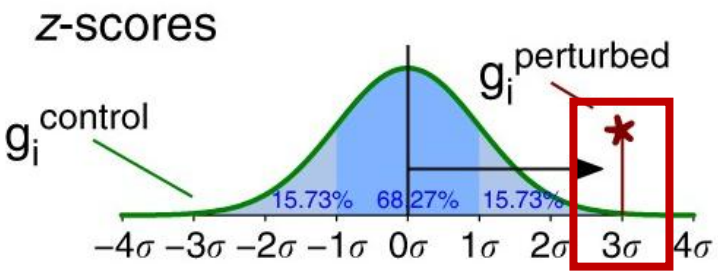
↓

For each gene per experiment:



Log-transformed expression

For gene X:

$$Z = P \times W$$

Gene X (Experiments)
$z_{x,1}$
$z_{x,2}$
…
$z_{x,n}$

n is the number of experiment

$1 \times n$

Z-score

11 pathways (Experiments)		
1	…	1
1	…	0
…	…	…
0	…	0

1 for activation, -1 for inhibition, otherwise 0

$11 \times n$

Pathway perturbation

Gene X (Pathways)
$w_{x,1}$
$w_{x,2}$
…
$w_{x,11}$

1×11

Weight

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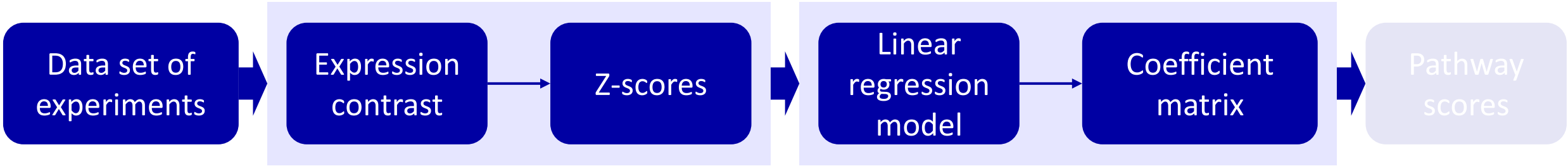

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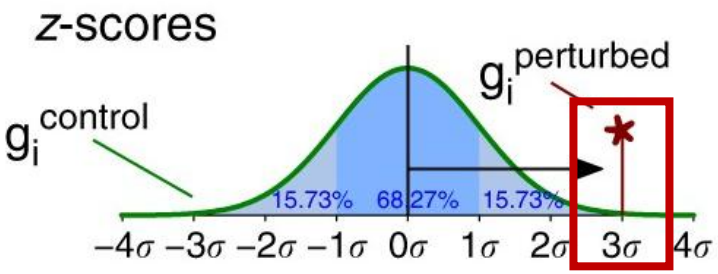
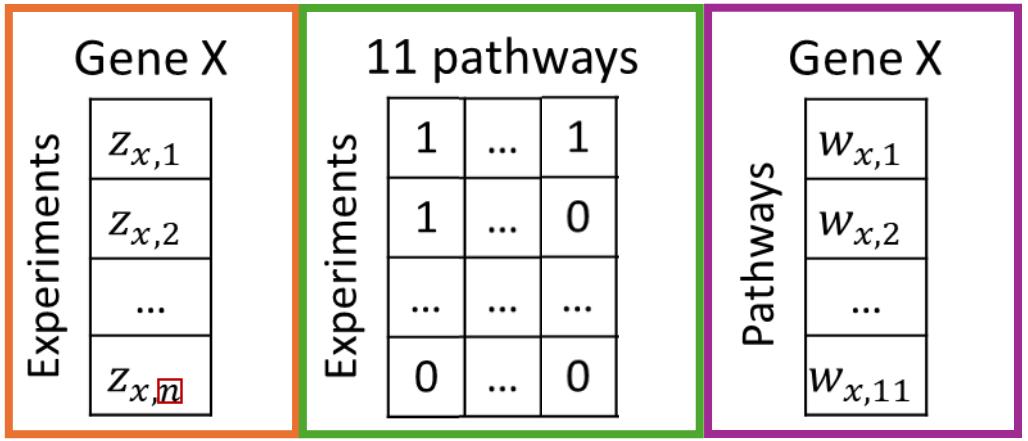


For gene X:

$$Z \sim P \ldots W$$



n is the number of experiment

1 for activation, -1 for inhibition, otherwise 0

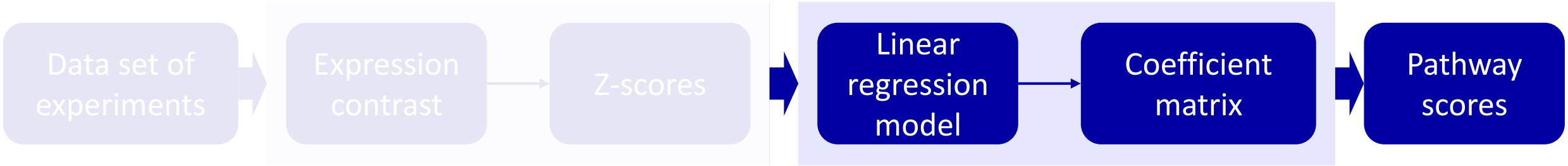
$1 \times n$ **Z-score**

$11 \times n$ **Pathway perturbation**

1×11 **Weight**

Fit for the weights for 11 pathways **per gene** → identify ~100 PRGs **per pathway**

PROGENy uses downstream gene signatures to reflect pathway activity

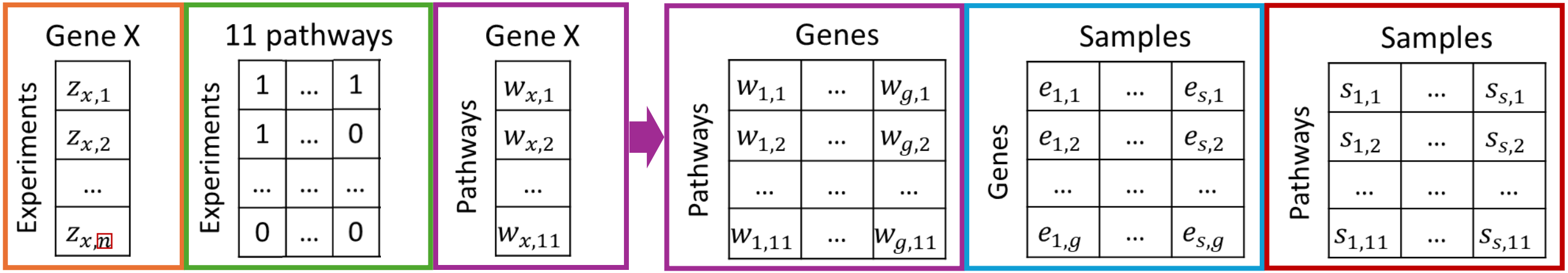


For gene X:

$Z \sim P \ldots W$

For samples:

$W \times E = S$



n is the number of experiment

1 for activation, -1 for inhibition, otherwise 0

g is the number of PRGs

s is the number of samples

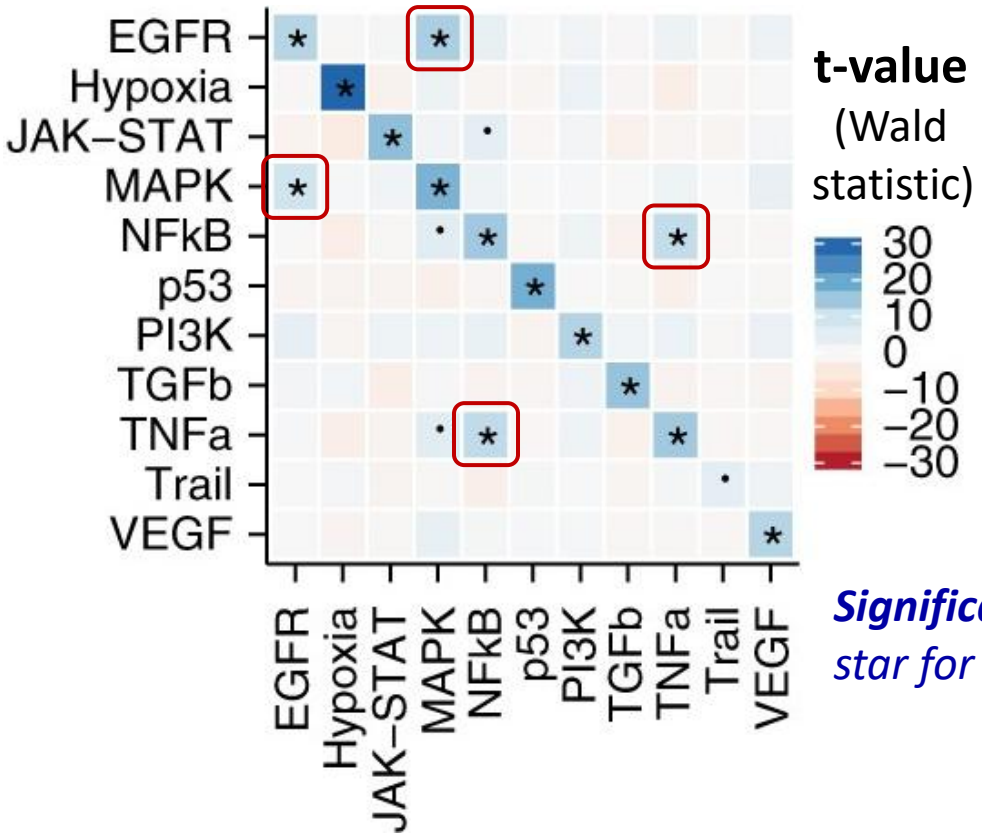
$1 \times n$	$11 \times n$	1×11	$g \times 11$	$s \times g$	$s \times 11$
Z-score	Pathway perturbation	Weight	Weight	Expression	Score

Fit the weights for 11 pathways **per gene** → identify ~100 PRGs **per pathway** → score 11 pathways **per sample**

PROGENy scores accurately reflect pathway activity

ANOVA

Pathway score



Pathway perturbation

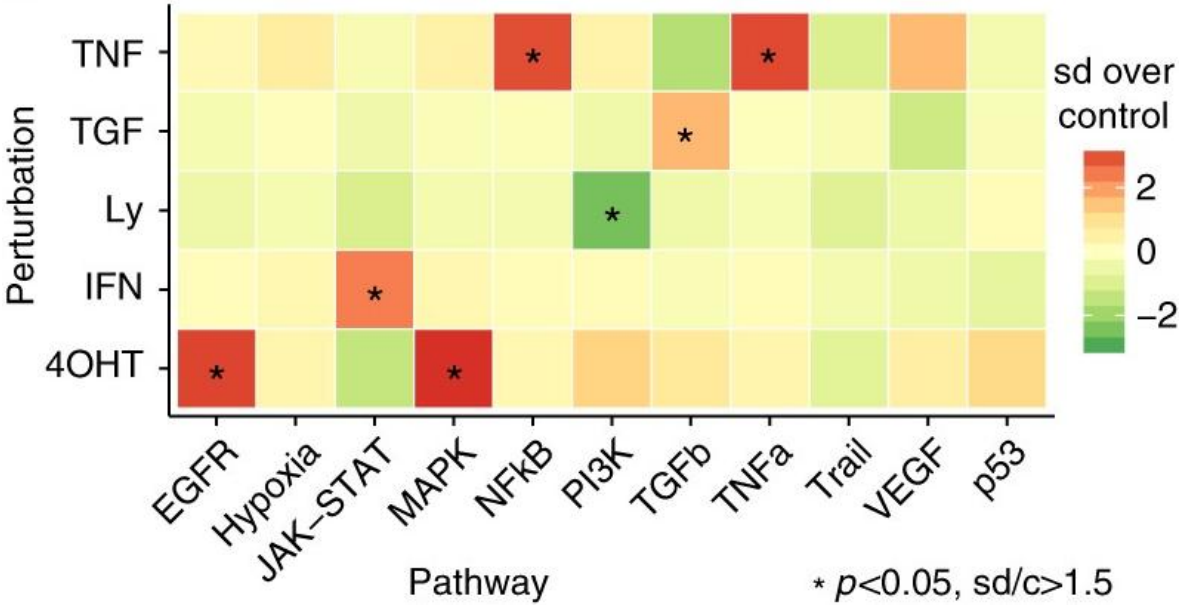
Leave-one-out strategy: train on all samples except one -> evaluate on the held-out sample

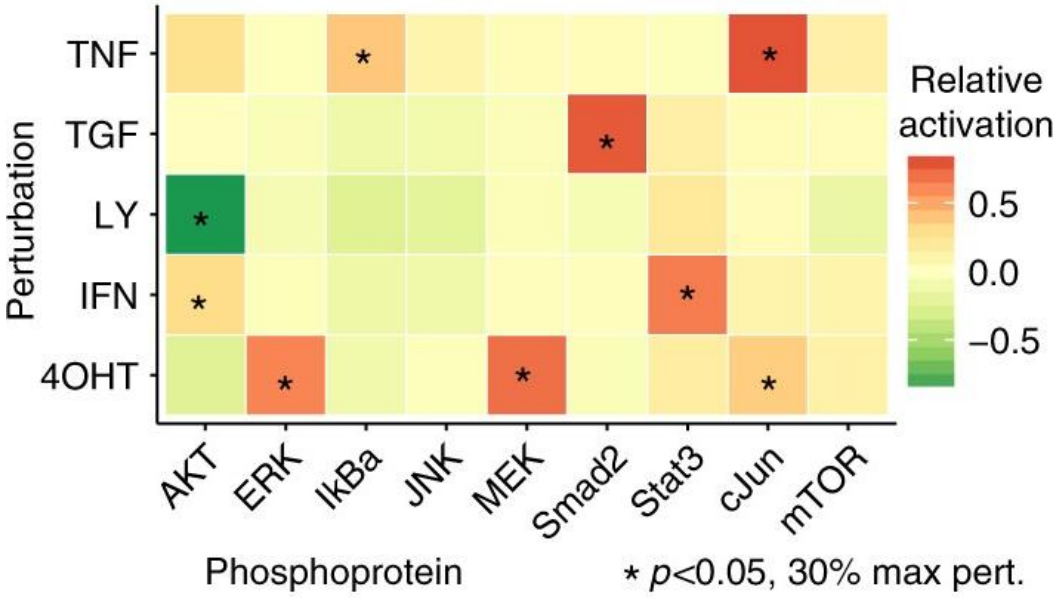
Significance: *dot for FDR* $< 10^{-5}$, *star for FDR* $< 10^{-10}$

- Pathway scores are strongly associated with corresponding perturbations
- Cross-talks are biologically plausible

PROGENy scores accurately reflect pathway activity

t-test





PROGENy correctly infers pathway activity in the **independent** HEK293 experiments

Perturbation experiments activate corresponding signaling proteins

Summary of Model Construction

Methodology

Perturb → learn signatures

From many perturbation experiments, fit **Z ~ P** to get pathway gene weights and identify ~100 PRGs per pathway

Score samples

Weighted sum of ~100 PRGs → 11 pathway scores per sample

Data

Data cleaning

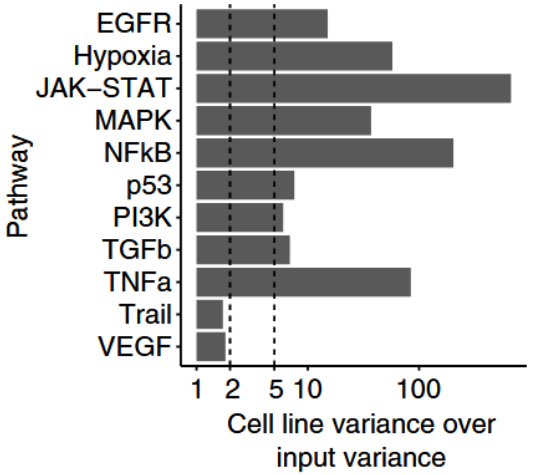
- Log transform + **z-score** standardization
- **Quality controls**
- **Filter** low-signal genes/samples
- **Align platforms** mixing arrays vs RNA-seq

Batch effects

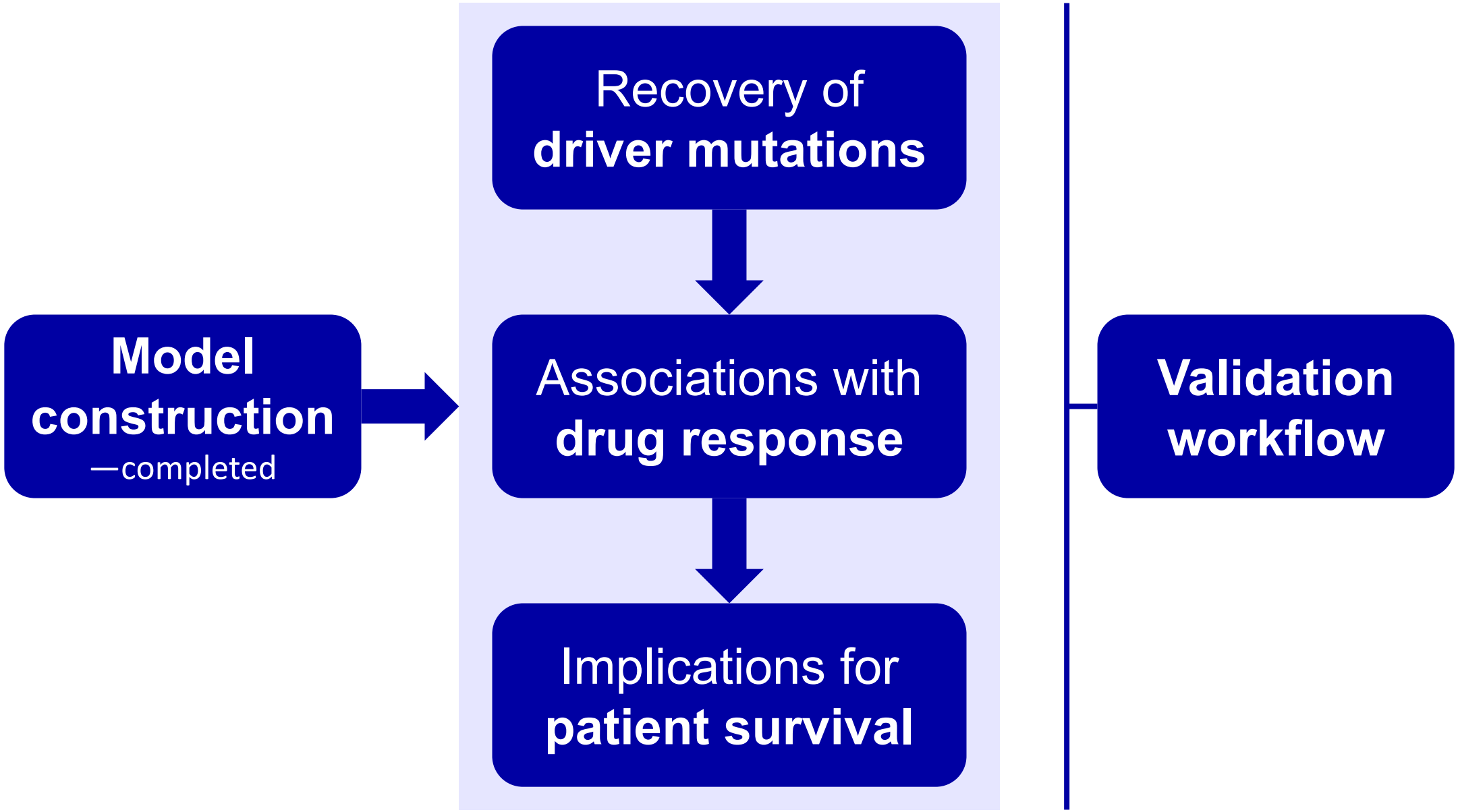
- **Detect** **PCA** -> correlation with lab/date/platform/…
- **Correct** Add **confounding factors** (batch/tissue)

Robustness checks

- **Leave-one-out** validations
- **Bootstrap** the training experiments



PCA: Principal Component Analysis


Model construction
—completed
Recovery of driver mutations
Associations with drug response
Implications for patient survival
Validation workflow

Model construction
—completed

Recovery of **driver mutations**

Associations with **drug response**

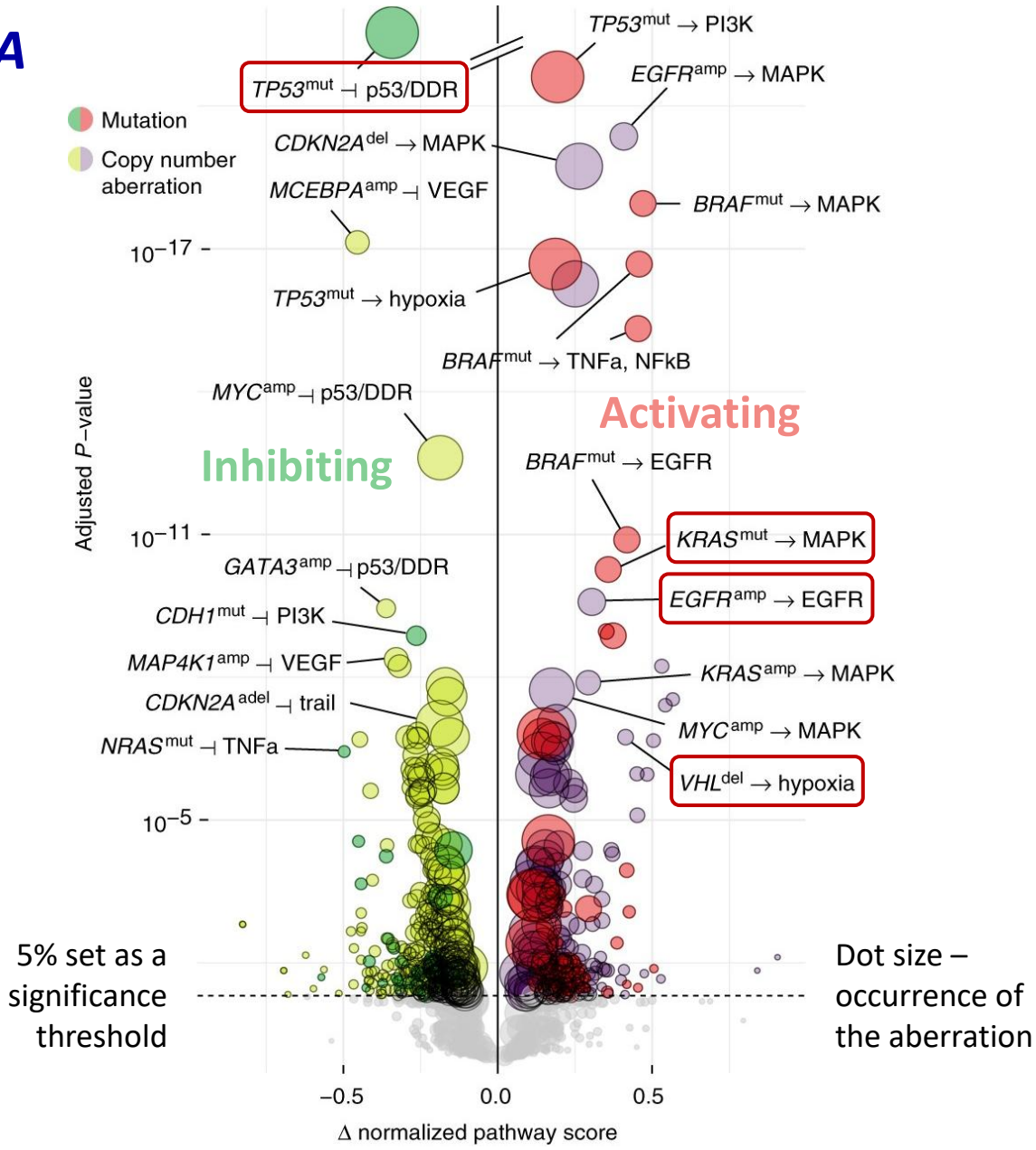
Implications for **patient survival**

Validation workflow

PROGENy recovers the effect of driver mutations

Cancer type	**Mutation status**	Survival	**Pathway score**

ANOVA



- **Pan-cancer** associations between driver mutations and pathway scores – *Data from the **TCGA** dataset*
- Pathway scores can distinguish between activating and inhibiting driver mutations

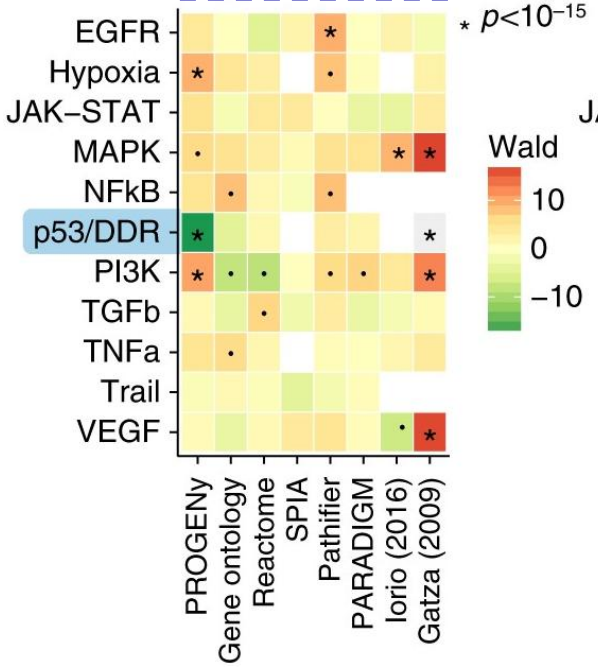
TCGA: The Cancer Genome Atlas

PROGENy recovers the effect of four well-known driver mutations

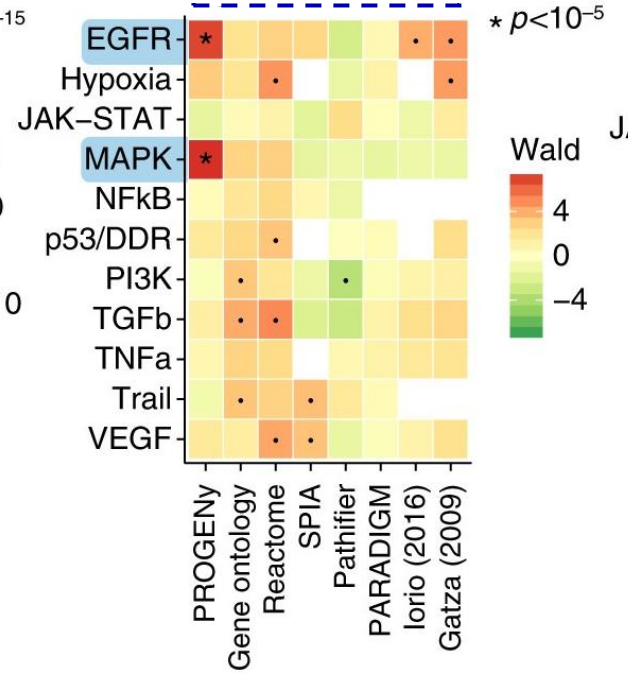
Cancer type	**Mutation status** mutated vs. wt	Survival	**Pathway score**

Pathway scores

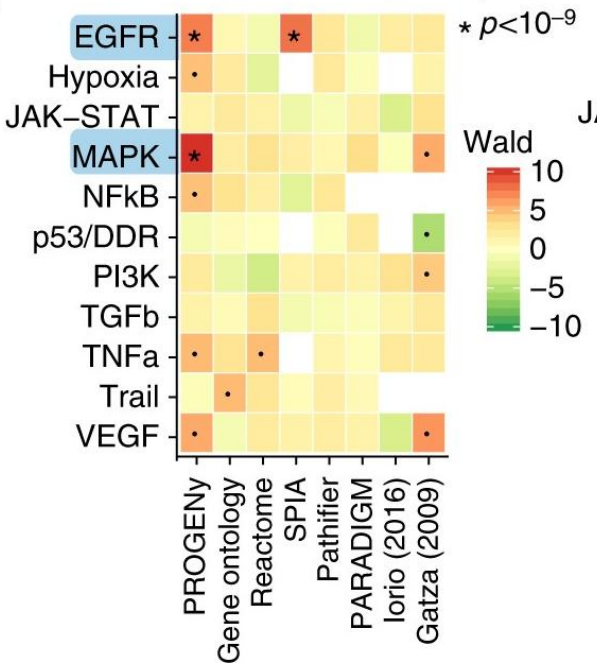
TP53-mutated



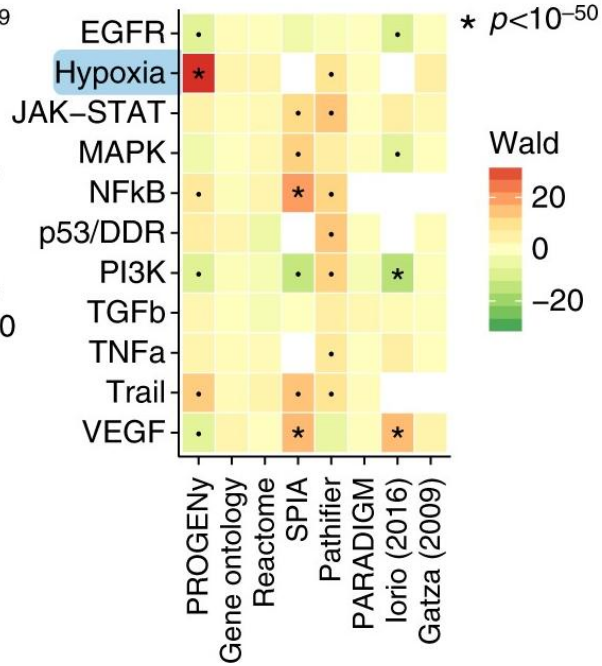
KRAS-mutated



EGFR-amplified

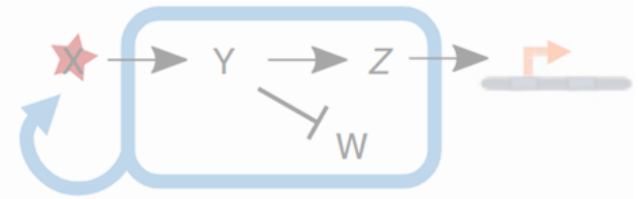


VHL-mutated

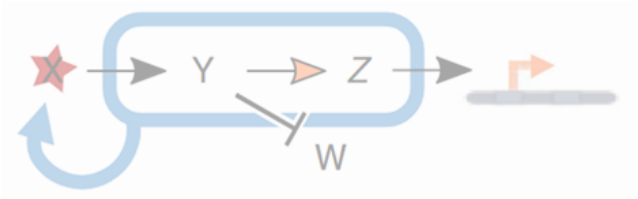


Different methods

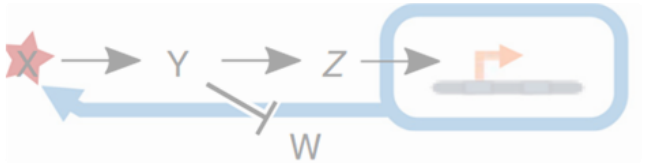
GO, Reactome

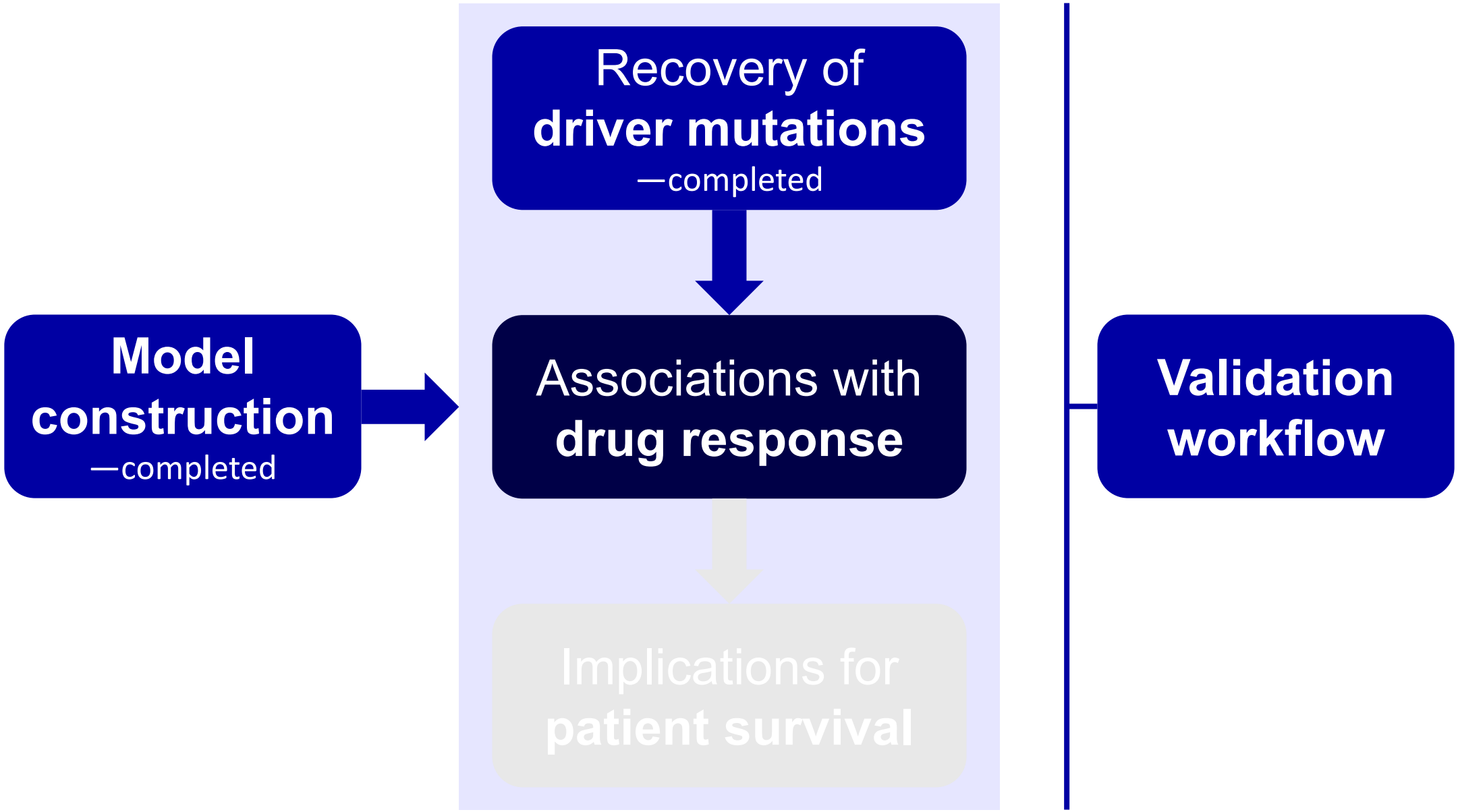


SPIA, Pathifier, PARADIGM



PROGENy, Iorio (2016), Gatza (2009)



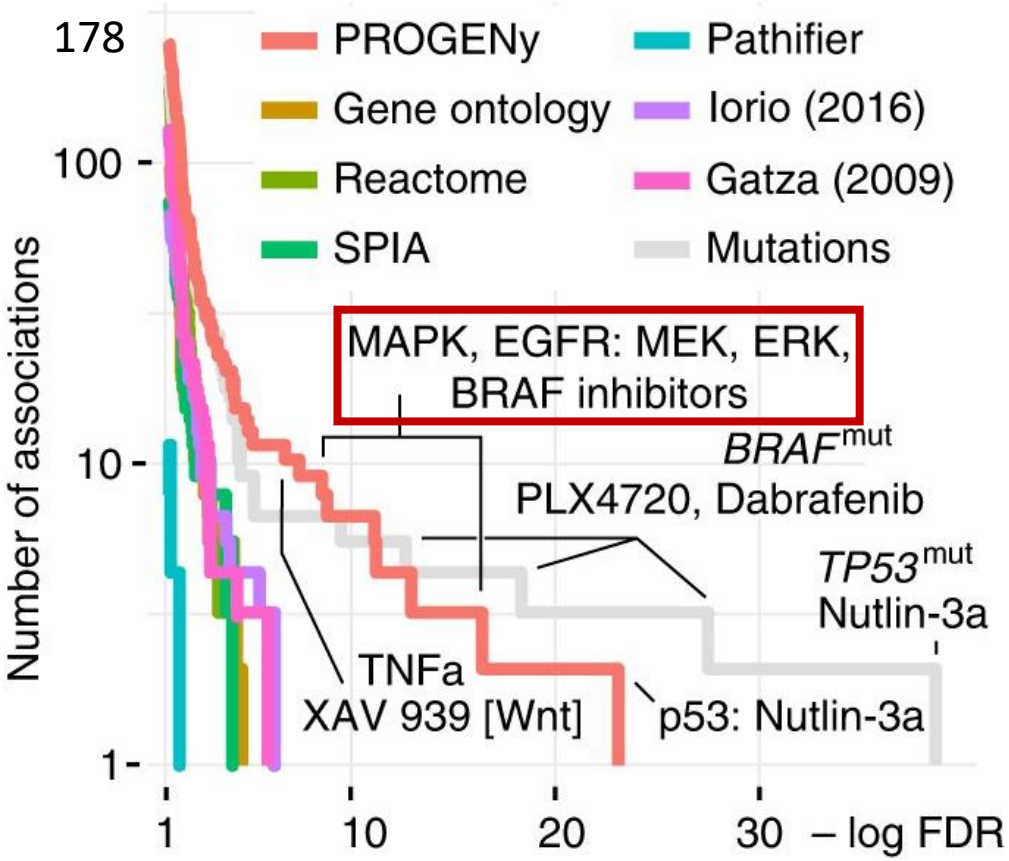

Recovery of
driver mutations
—completed
Model
construction
—completed
Associations with
drug response
Implications for
patient survival
Validation
workflow

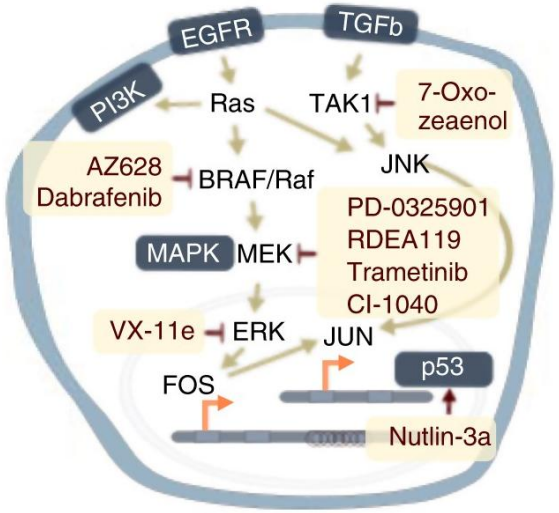
PROGENy produces more and stronger associations with drug response

Cell line – 805 cell lines	Mutation status	IC_{50} – 265 drugs	**Pathway score**

ANOVA

Test the correlation between **drug sensitivity** (IC_{50}) of 265 drugs and predicted 11 **pathway scores** across 805 cell lines – *Data from the **GDSC** project*

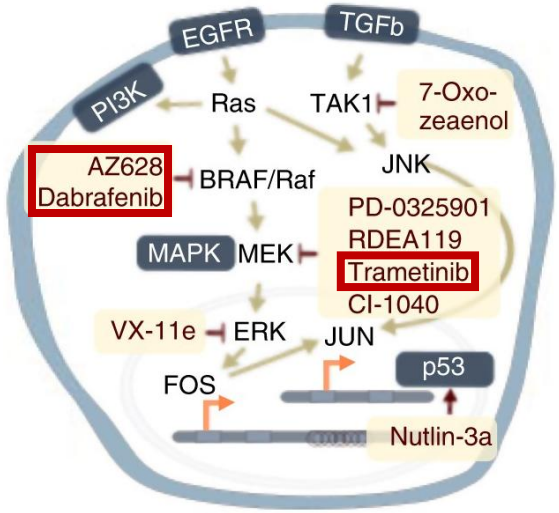


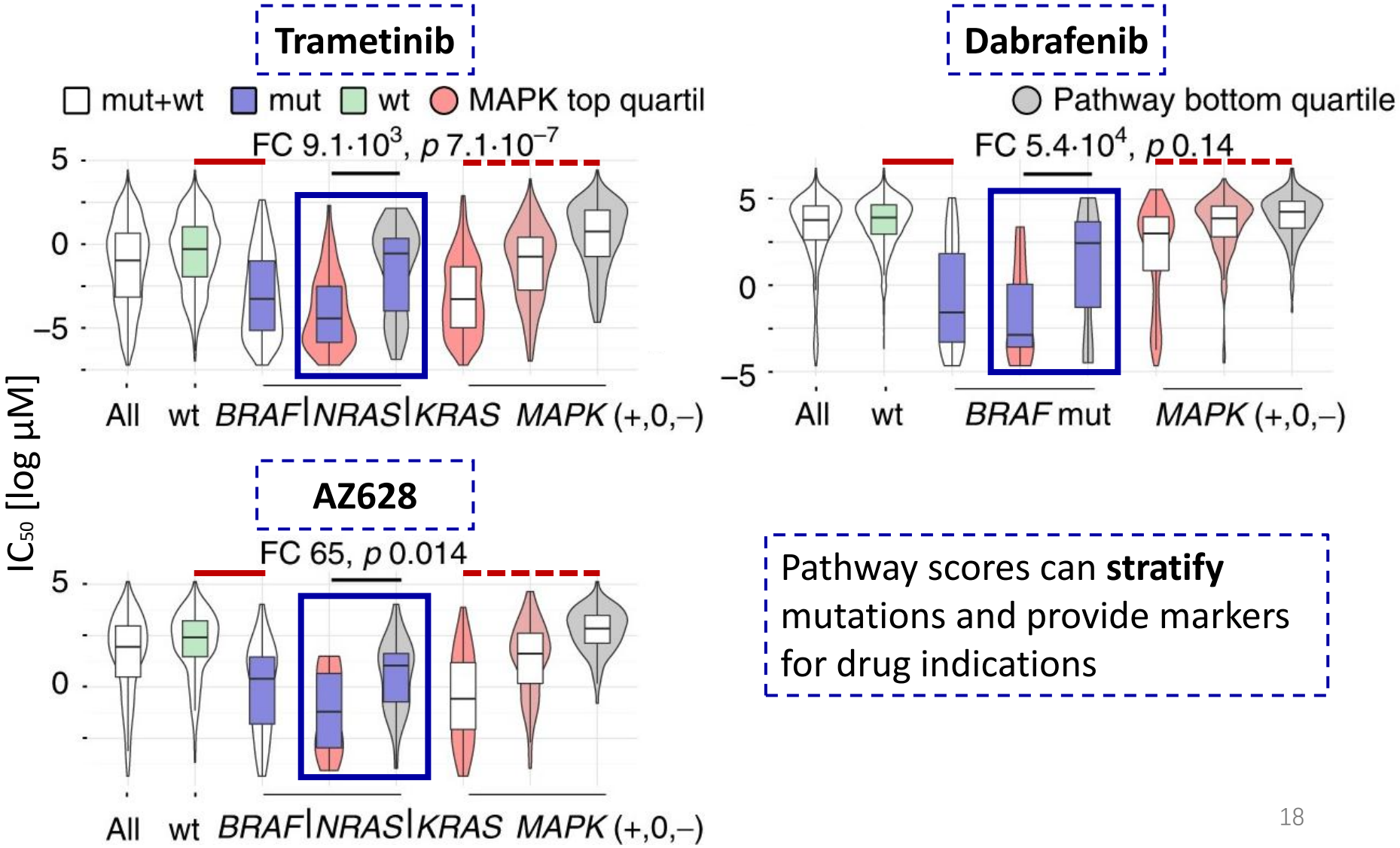


GDSC: Genomics of Drug Sensitivity in Cancer
FDR: False Discovery Rate

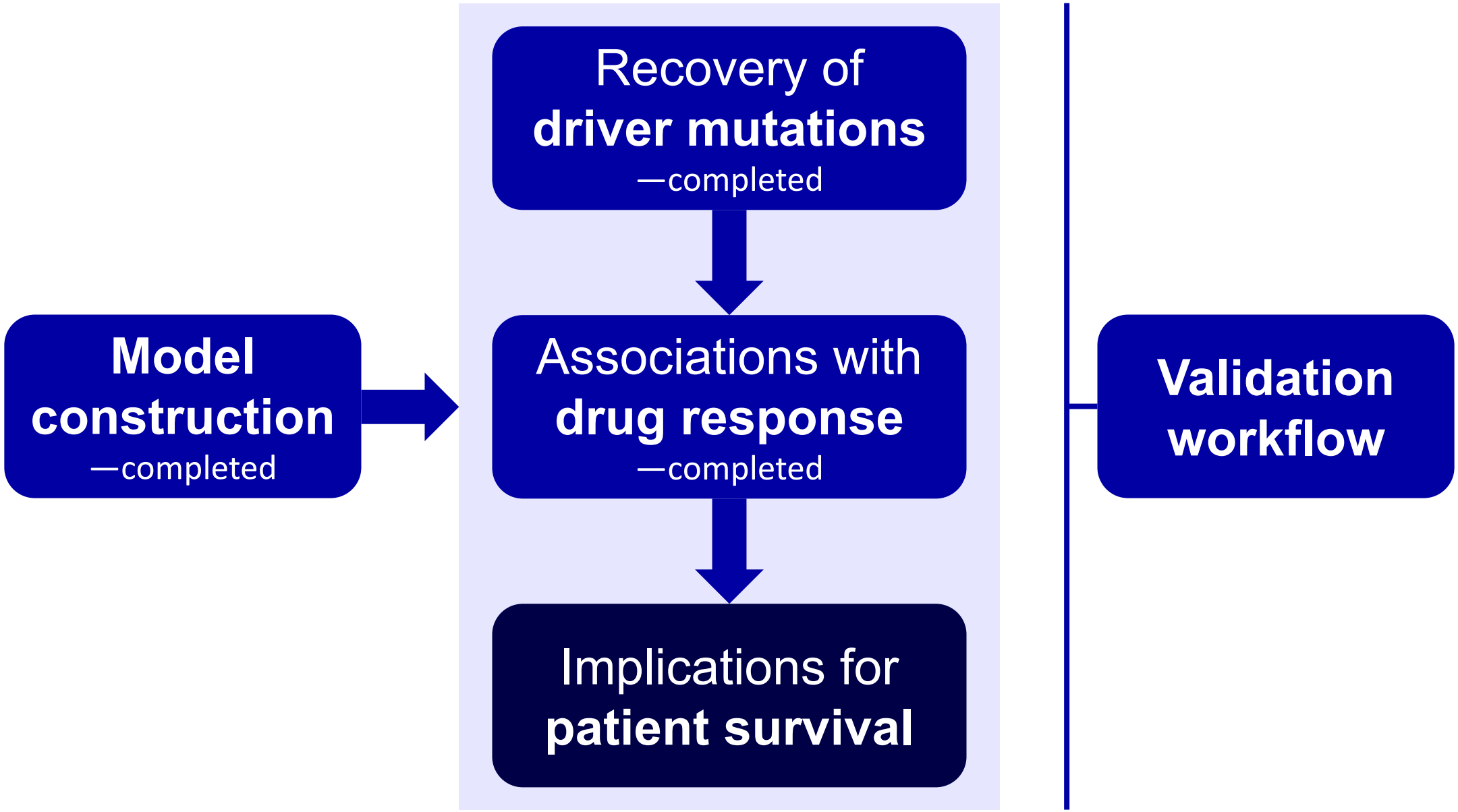
MAPK score associates drug response across cancer types

Cell line – 805 cell lines	**Mutation status**	**IC_{50} – 265 drugs**	**Pathway score**





Pathway scores can **stratify** mutations and provide markers for drug indications


Model construction
—completed
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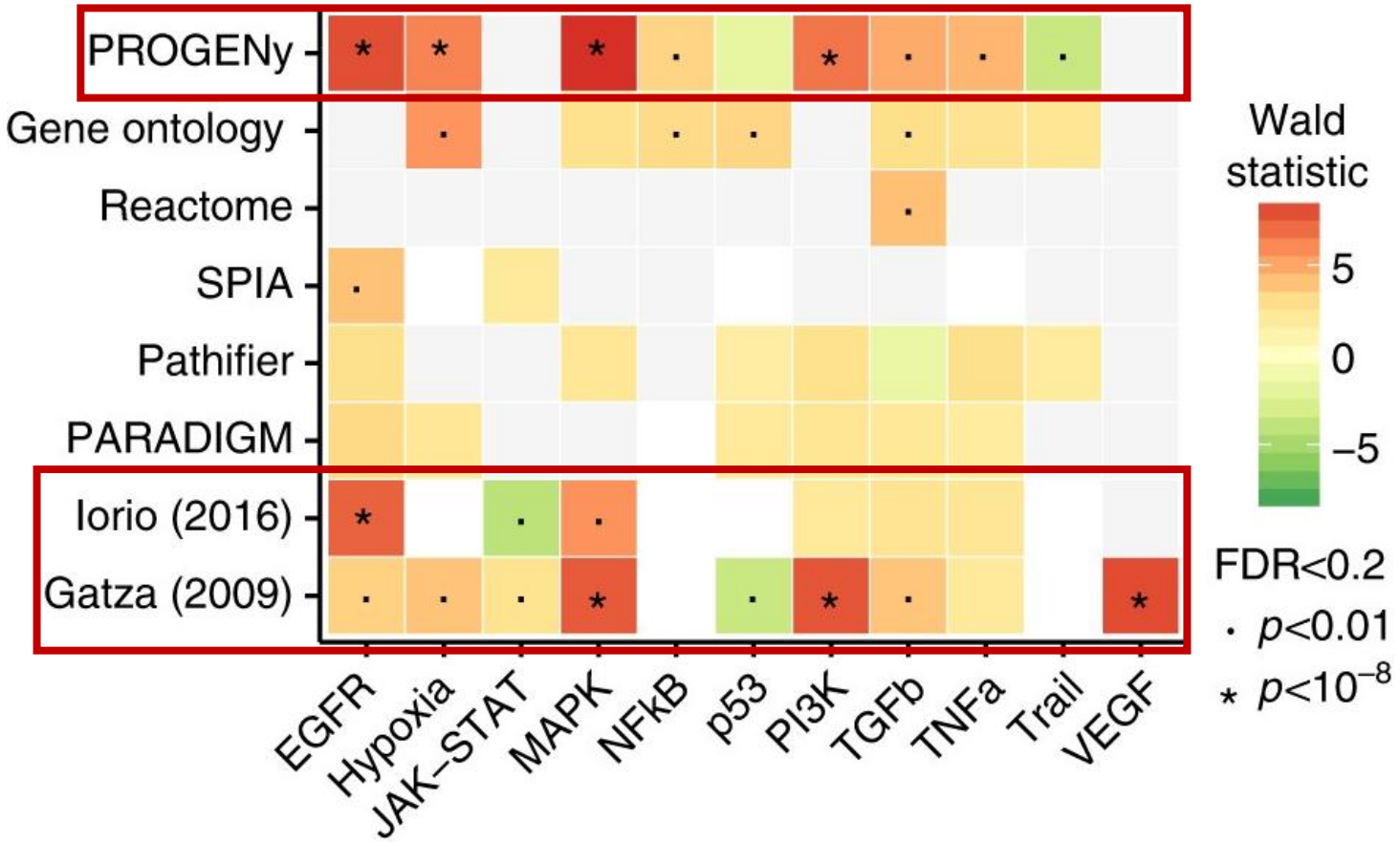
Associations with drug response
—completed
Implications for patient survival
Validation workflow

PROGENy produces more associations with patient survival

Cancer type	Mutation status	**Survival**	**Pathway score**

ANOVA

Different methods



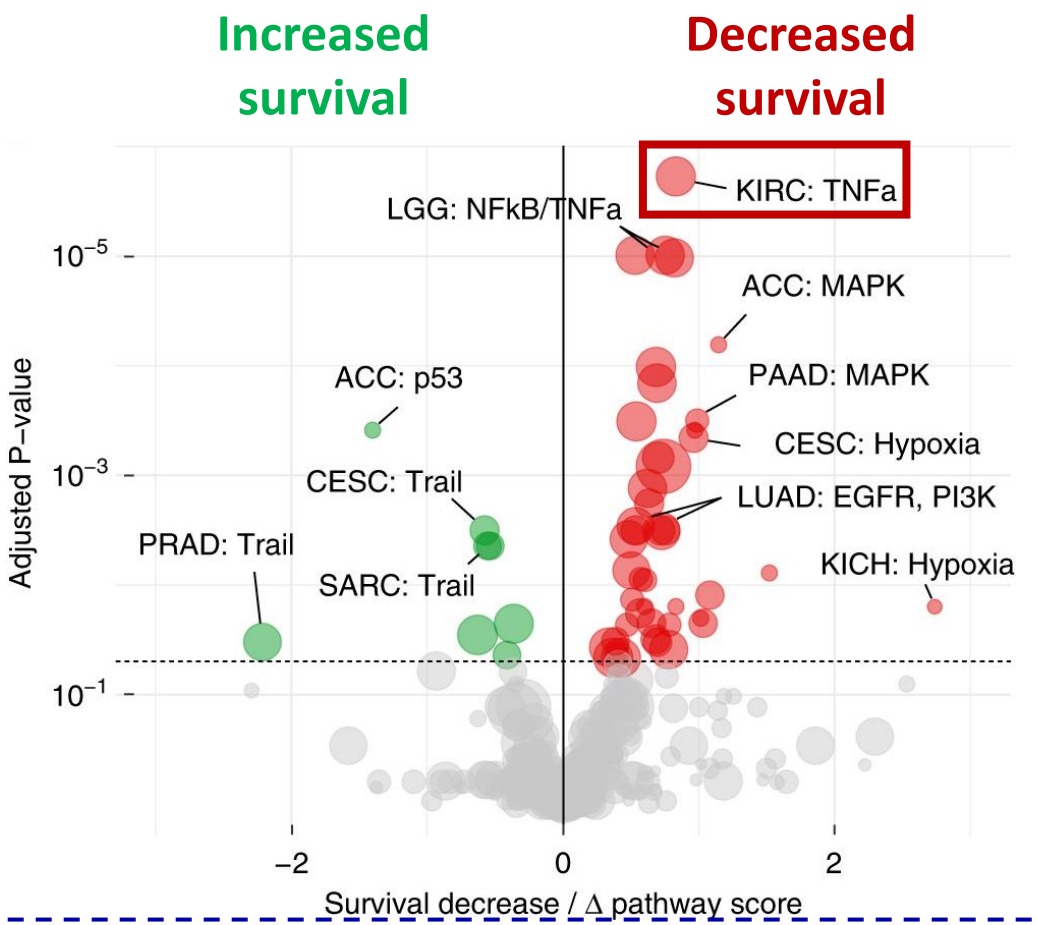
Pan-cancer associations

- Signature-based methods generally outperform pathway mapping for survival associations

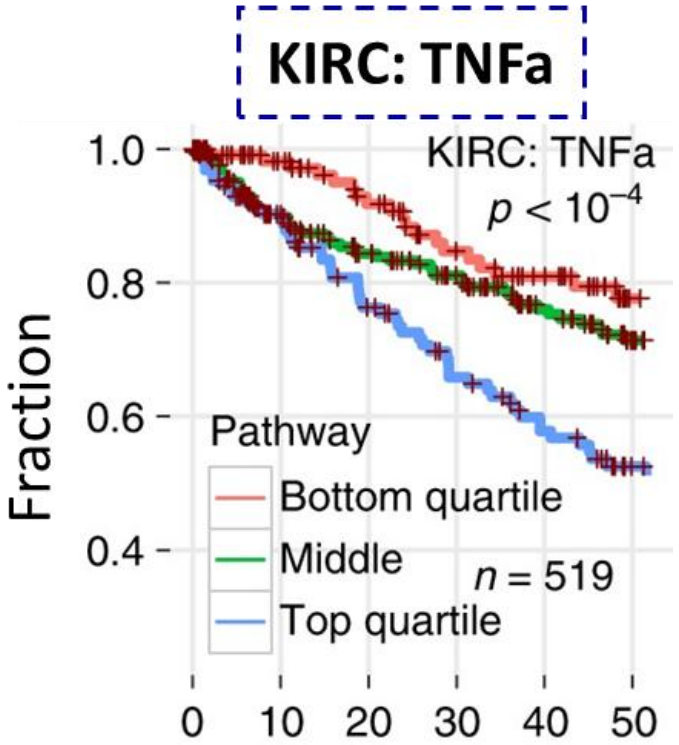
PROGENy produces more associations with patient survival

Cancer type	Mutation status	Survival	Pathway score

Cancer-specific associations

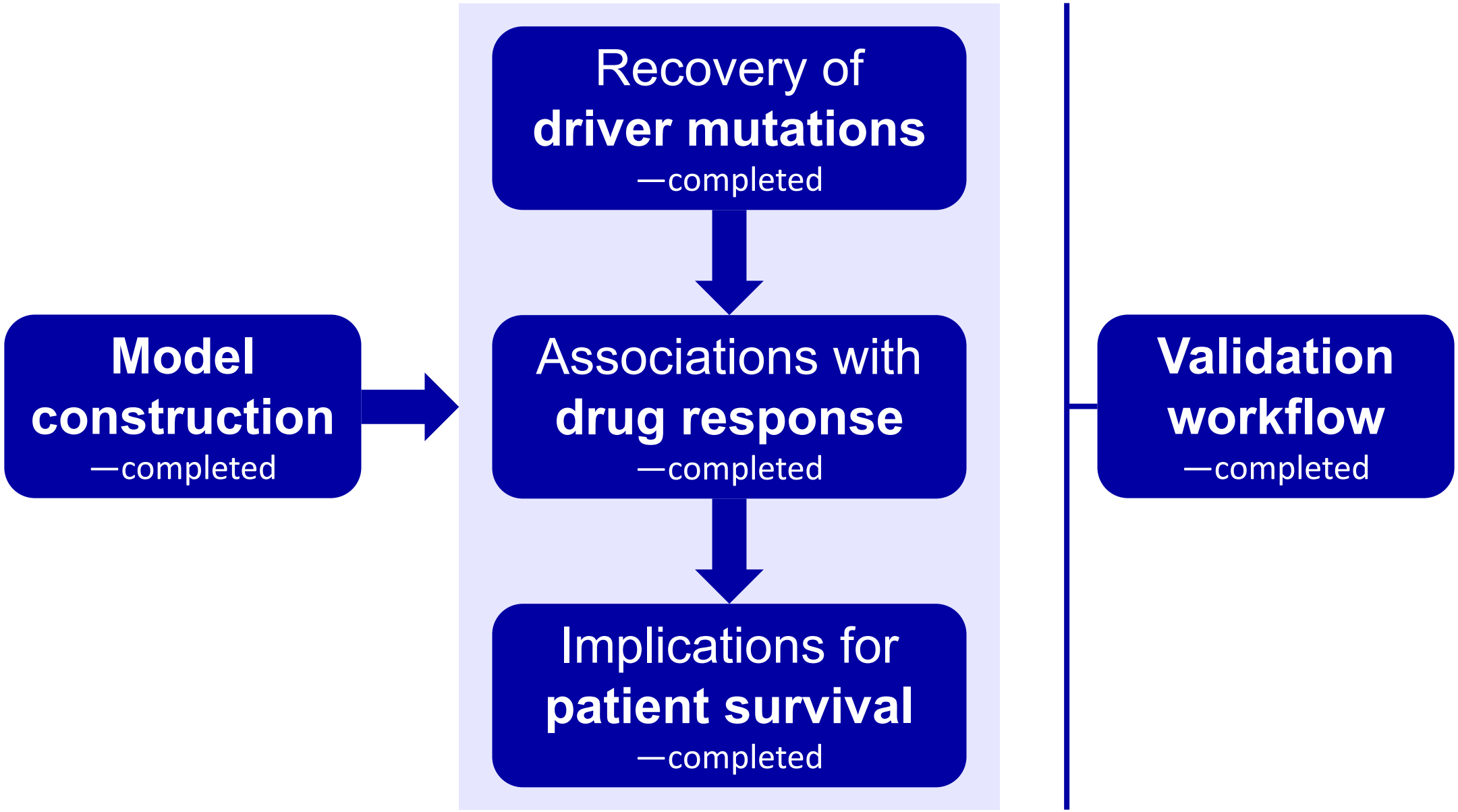


Pathway scores can distinguish oncogenic and tumor suppressor pathways



Pathway scores can **stratify** patients in terms of one year survival

KIRC: kidney renal carcinoma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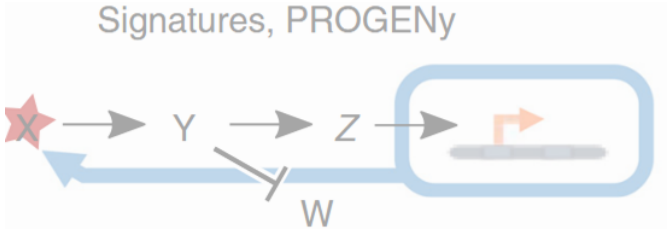
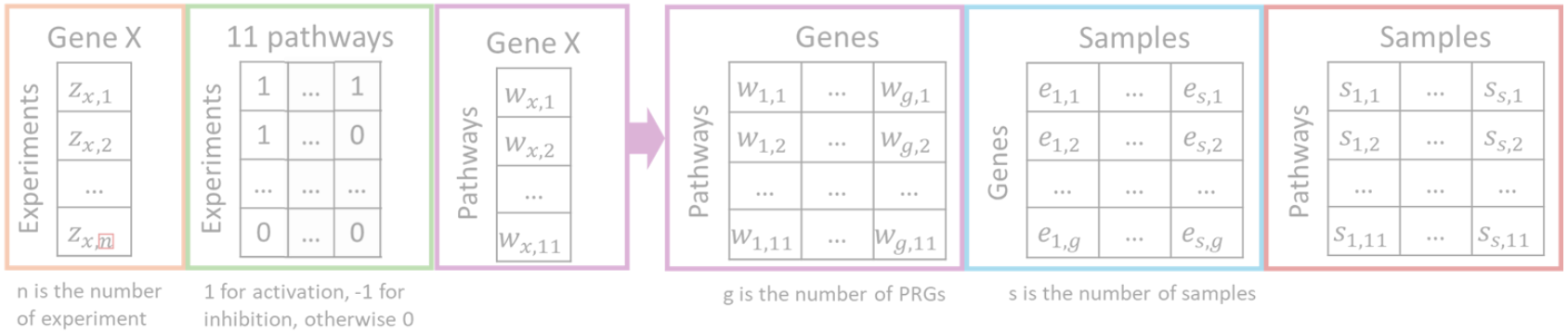
Problem

- Read **pathway activity** from gene expression
- mRNA of pathway members ≠ activity (post-translational control)



Method

- Learn **perturbation-response gene signatures** per pathway
- From activate/inhibit experiments via **simple regression**

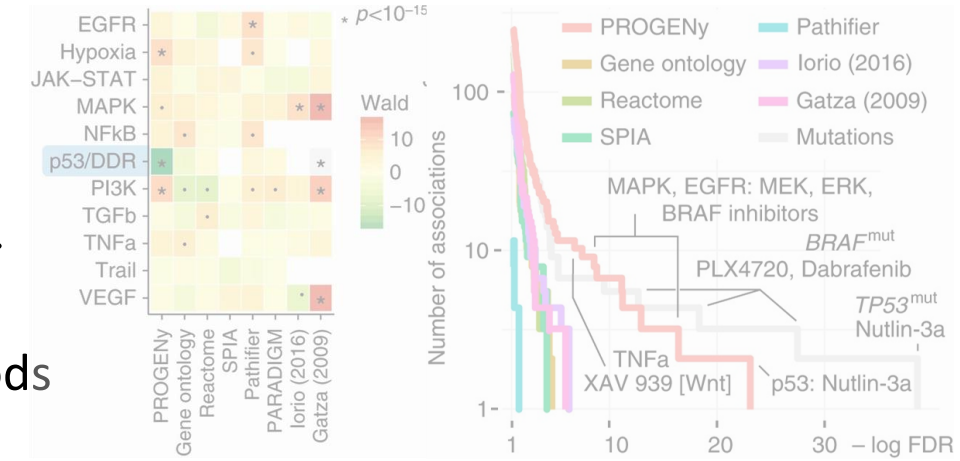


Scoring

- **Weighted sum** → one activity score per pathway/sample
- Works on tissues and cell lines (bulk/single-cell RNA-seq)

Validation

- **Perturbations → mutations → drugs → survival**
- Overperforms mapping methods



PROGENy provides **open pathway signatures** that work across datasets and are easy to **plug into any RNA-seq matrix**

Thank you!

HP-F2: Part 3—Bioinformatics

Yilu Wang
Major Cancer Biology
04.12.2025